Day 1: Python Variables, Types, Operators

Oct 19 to 20, 2026 | Python track | Revision notes

1. Variables

- A variable is a **name tag** stuck on a value. `num = 85` puts the tag `num` on the value 85.
- = means **store**, not 'equal to'. The tag can move: `num = "pass"` is allowed (dynamic typing). The type belongs to the value, not the name.
- **Interview edge:** Python variables are references, not containers. `id(num)` changes after reassignment, which shows the tag moved to another object.

2. Types and input()

- Main types: `int` (85), `float` (85.5), `str` ("85"), `bool` (True/False). Check with `type()`.
- `"85" + "5"` gives `"855"` (text joins). `"85" + 5` raises TypeError.
- `input()` **always returns text**. Convert it: `marks = int(input("Enter marks: "))`.
- **Interview edge:** `int("85.5")` raises ValueError (use `int(float("85.5"))`). `int(85.9)` gives 85: it cuts, it does not round. `"7" * 3` gives `"777"`.

3. Operators

Operator	Example	Result	Meaning
/	85 / 2	42.5	True division. Always returns a float (4 / 2 is 2.0)
//	85 // 2	42	Floor division: rounds down. -85 // 2 is -43, not -42
%	85 % 2	1	Remainder. 0 means even. -7 % 3 is 2 in Python
==	num == 85	True	Asks 'is it equal?'. One = stores, two == compare
and / or	marks>=40 and att>=75	needs both / either	and: both True. or: one is enough

- **Interview edge:** `0.1 + 0.2 == 0.3` is **False**. Floats are stored in binary, so some decimals are not exact (0.30000000000000004). For money use `round(x, 2)`.
- **Interview edge:** `and` / `or` short-circuit: `x != 0 and 10 / x > 2` never crashes at x = 0. `0 or "default"` returns `"default"`.
- `7 == 7.0` is True because == compares values, not types. `7 == "7"` is False.

4. Practice problems (redo from a blank file)

- Swap two variables without a third variable.
- Read a price with `input()`, add 18% GST, print the total (try 250.50 and 19.99).
- Print the type and value of `7 / 2`, `7 // 2`, `7 % 2`, `"7" + "2"`, `7 == 7.0` and write one line explaining each.

5. Recall questions (answer from memory, then check)

- What does it mean that a variable is a reference?
- Why does `input()` need int() or float()?
- What is the difference between /, // and %, and what does -7 // 2 give?
- Why is 0.1 + 0.2 == 0.3 False, and what do you do for money?